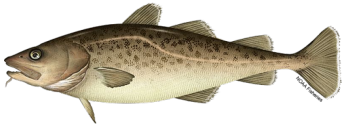
From heatwaves to recovery:

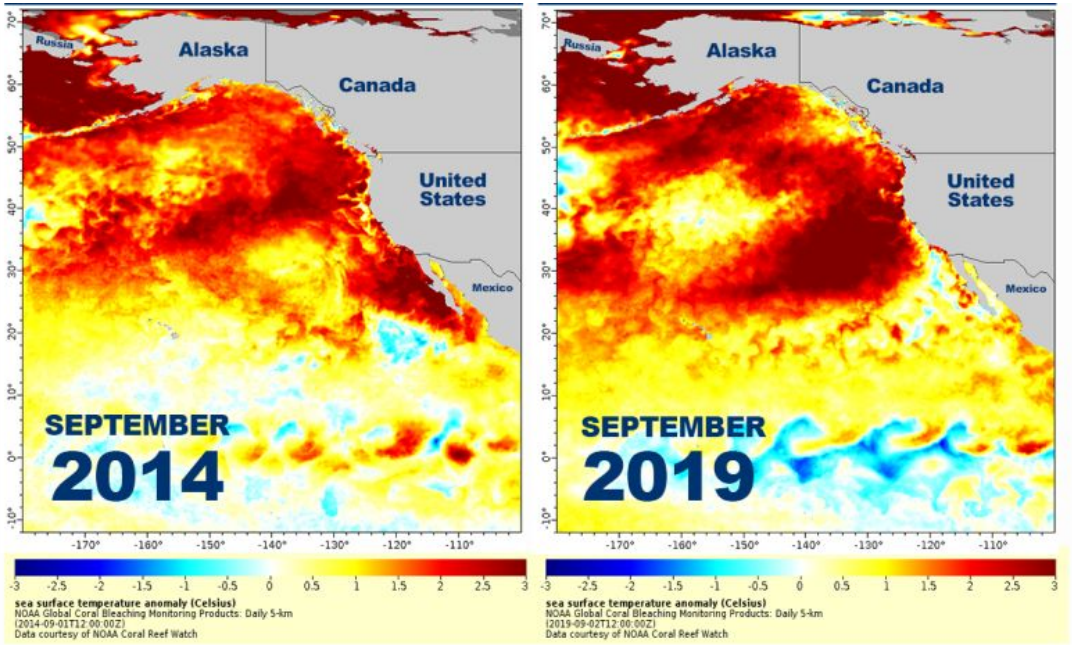
Tracking genomic signals of juvenile fitness traits in a Pacific cod nursery



Laura Spencer, NOAA AFSC Affiliate at the Univ. of Washington

Ben Laurel, NOAA AFSC Newport, OR
Louise Copeman, NOAA AFSC Newport, OR
Mary Beth Rew-Hicks, NOAA AFSC Newport, OR
Ingrid Spies, NOAA AFSC Seattle, WA
Steven Roberts, University of Washington SAFS

Heatwaves!



[NOAA feature story](), *September 05, 2019*



Pacific cod, *Gadus macrocephalus*

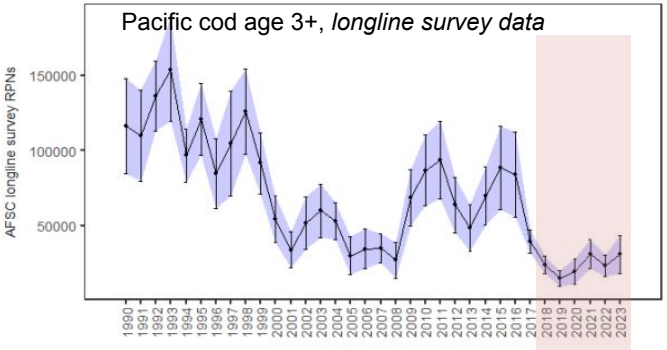
Extremely low cod numbers lead feds to close the Gulf of Alaska fishery for the first time

Alaska Public Media
Published December 6, 2019 at 8:22 PM AKST



Pacific cod (Photo Courtesy of Holland Dotts & the Alaska Marine Conservation Council)

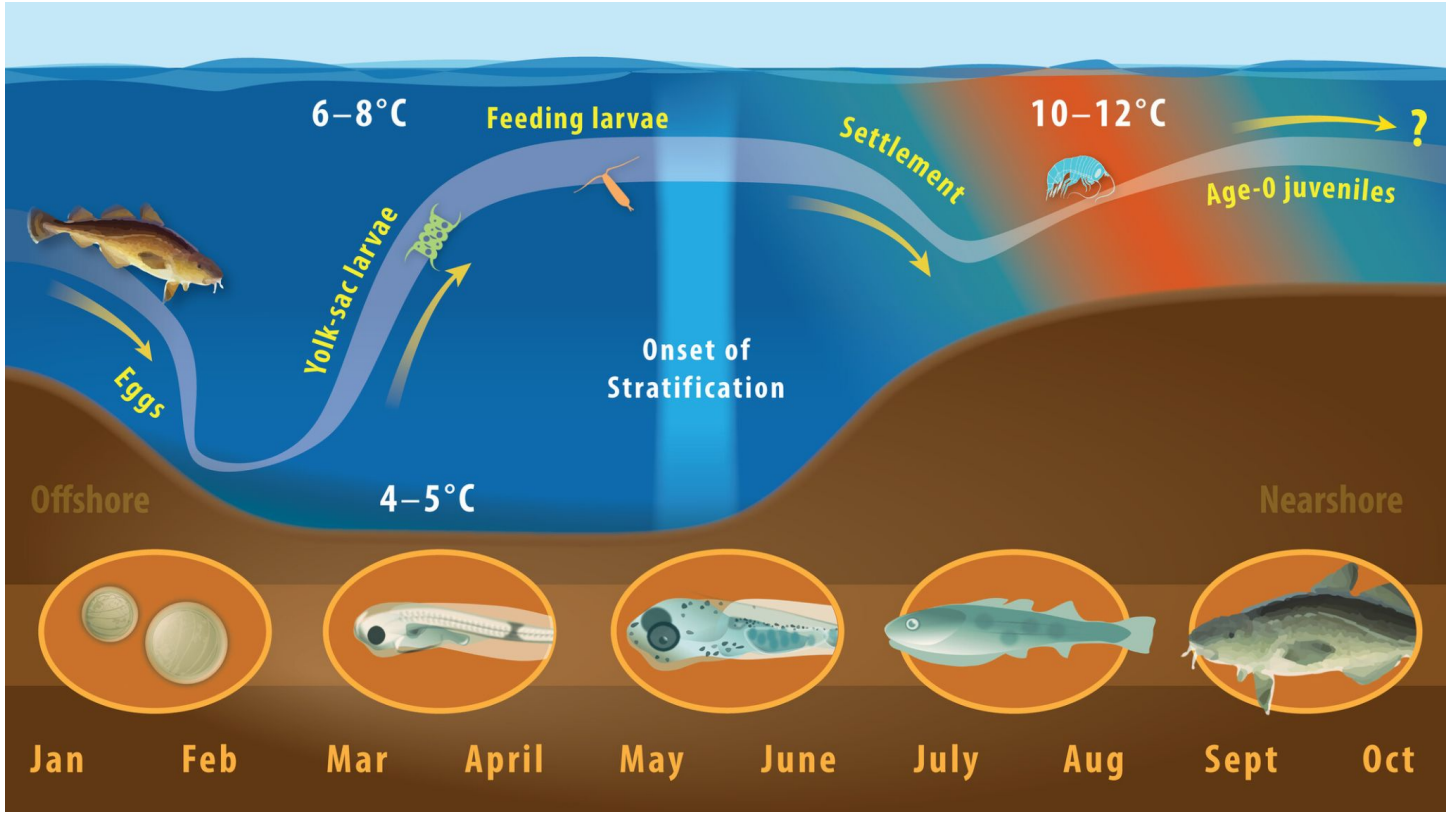
[Alaska Public Media](), *December 06, 2019*



[Stock Assessment Report, 2023, Gulf of Alaska](),



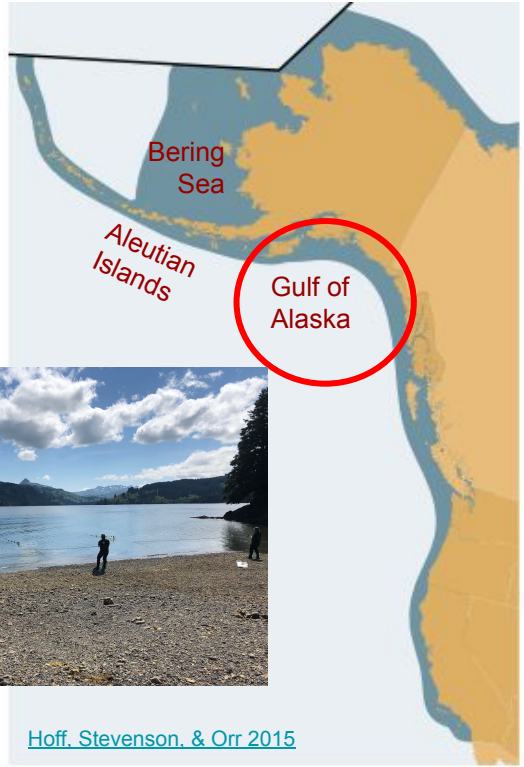
Pacific cod early life stages are very sensitive to temperature



Laurel et al. 2023, *in Fish & Fisheries*



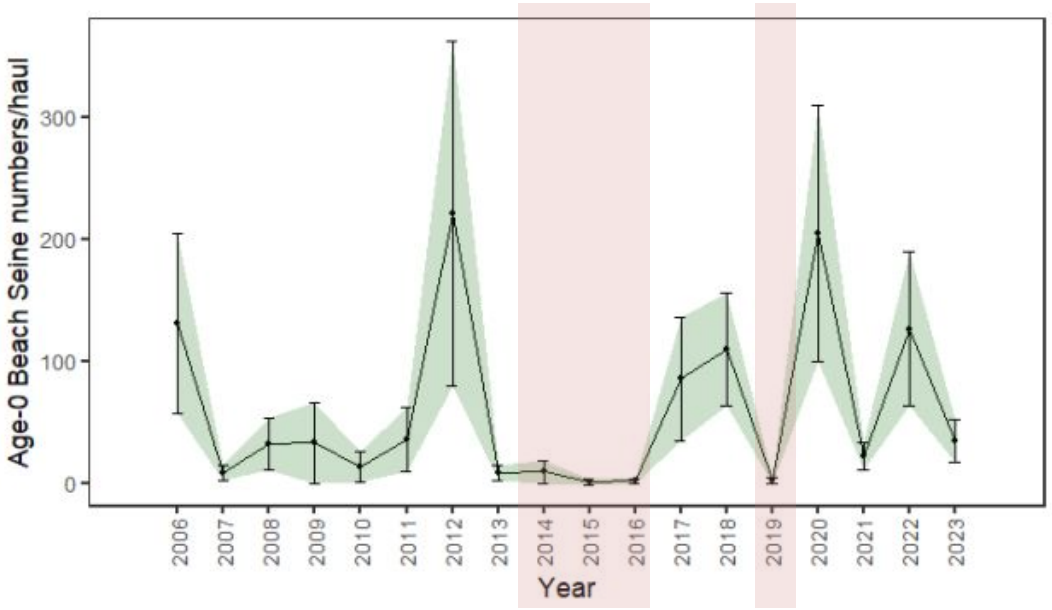
Low abundances likely related to low recruitment during heat waves





Hoff, Stevenson, & Orr 2015

Pacific cod juvenile (age-0) *beach seine survey CPUE*



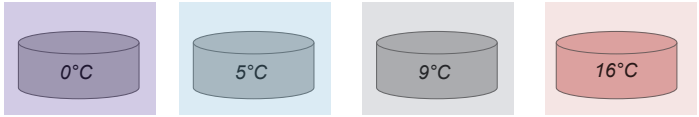
Stock Assessment Report, 2023, Gulf of Alaska,



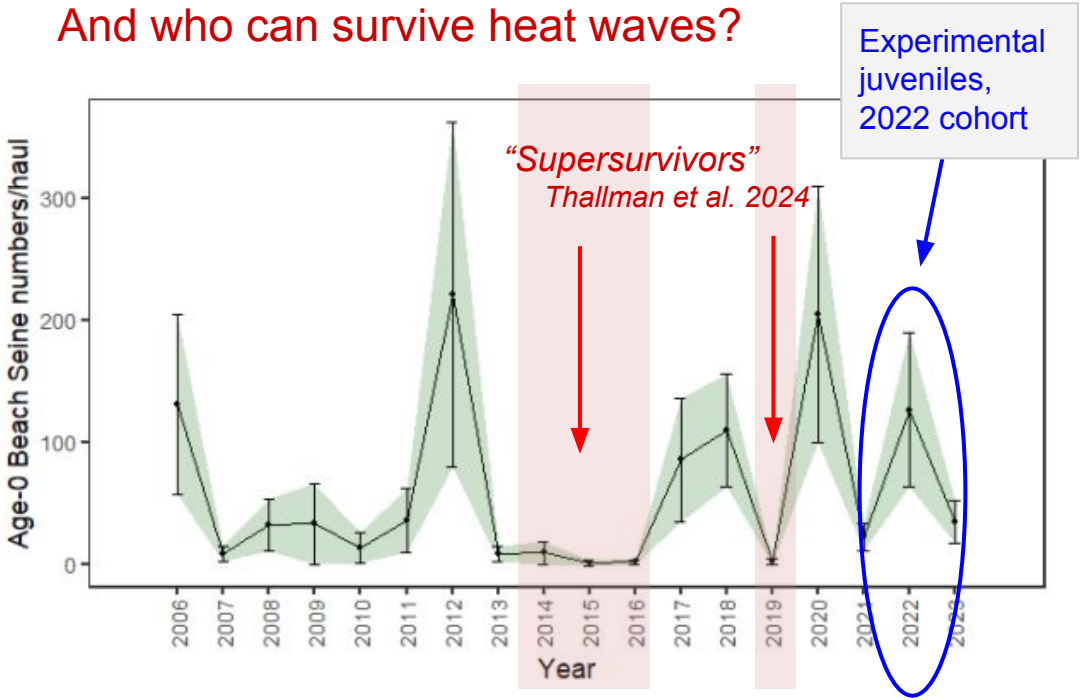
Question: How does temperature affect juvenile Pacific cod?

Juvenile Temperature Experiment





And who can survive heat waves?



Stock Assessment Report, 2023, Gulf of Alaska,



“Genome to Phenome” dataset across temperatures for Pacific cod





→ Individuals were tagged ←

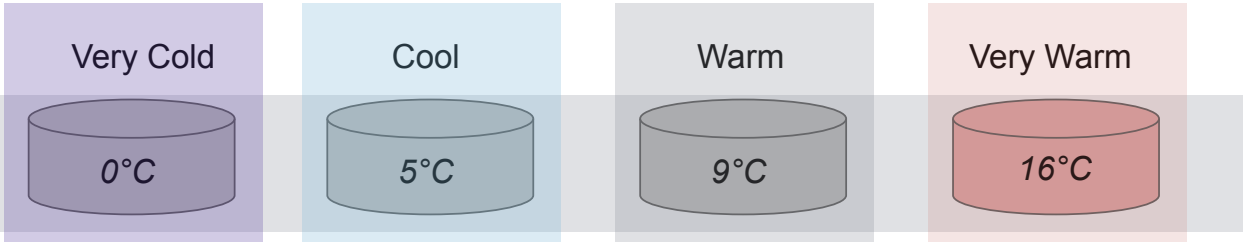
a. Genetics (lcWGS)

b. Gene expression in liver, n=18/temp (RNASeq)

c. Fitness Traits
   - Growth rates
   - Liver & Body Condition
   - Liver lipid concentrations (n=25/temp)
   - Survival



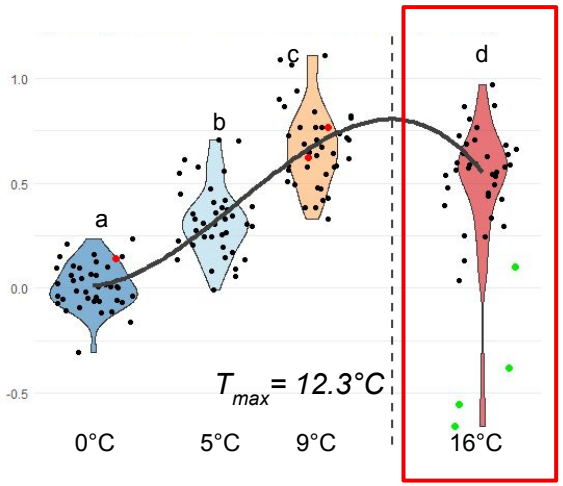
n=40 / temp.
4 week treatment



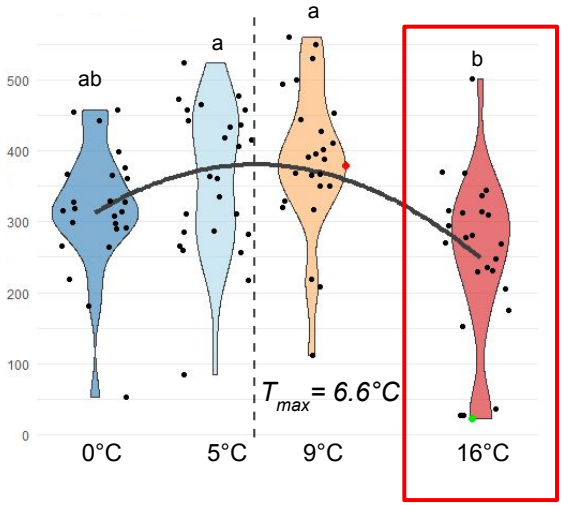


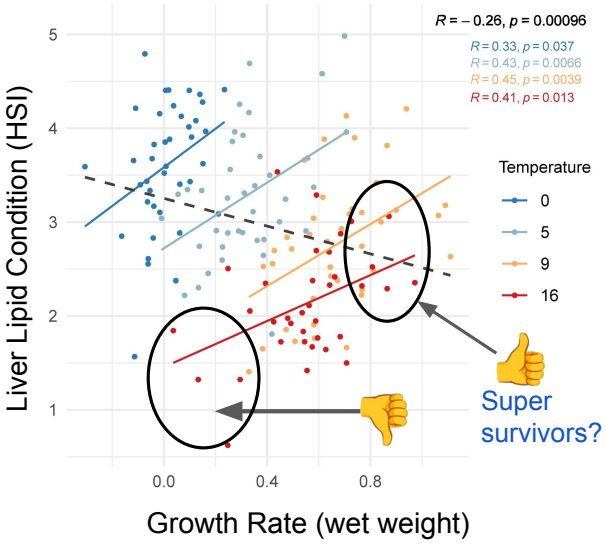
Warming slows growth, reduces energy reserves (lipid), & increases mortality - but growth & lipids are positively correlated

Growth Rate (weight, %/day)



Lipid Content in liver (mg)

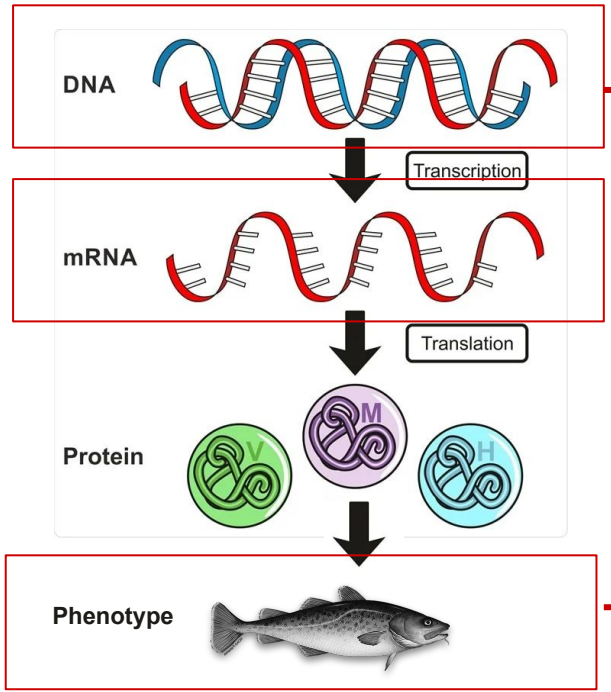




n=40/temperature

Published soon in GCB-Comm

“Genome to Phenome” dataset across temperatures



Adapted from: udaix/Shutterstock.com

→ Individuals were tagged ←

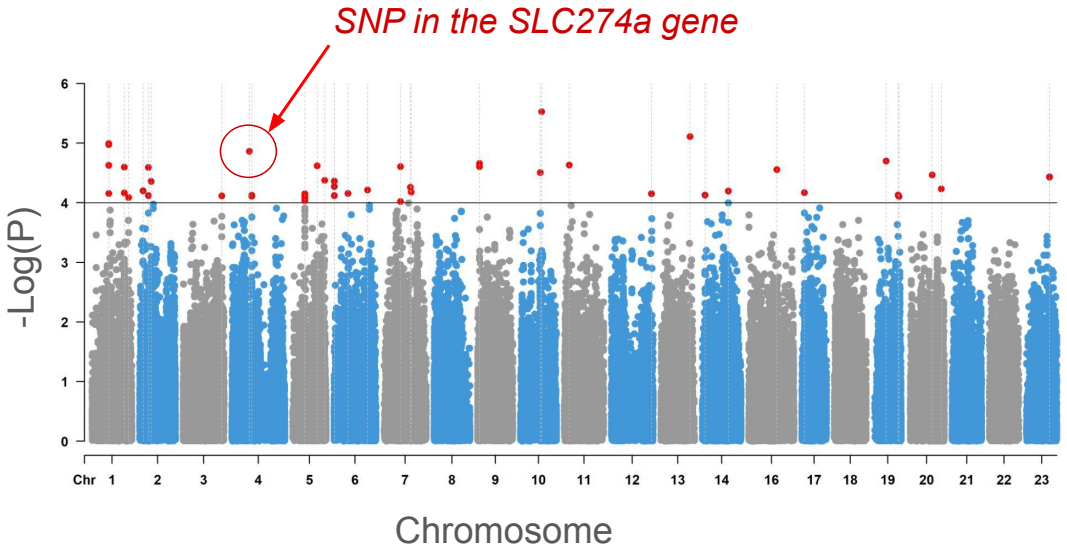
a. Genetics (lcWGS)
b. Gene expression in liver (RNASeq)
c. Fitness Traits
   - Growth rate
   - Liver & Body Condition
   - Liver lipid concentration, content
   - Survival

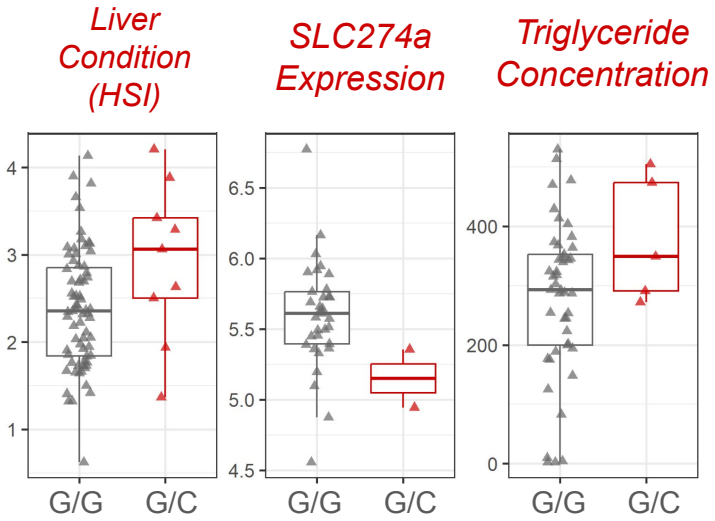
Which **sites on genome** are associated with **fitness traits?**

Genome-Wide Association Studies (GWAS) using single nucleotide polymorphisms (SNPs)



SNP markers putatively associated with **Liver Condition (HSI)** in Pacific cod juveniles



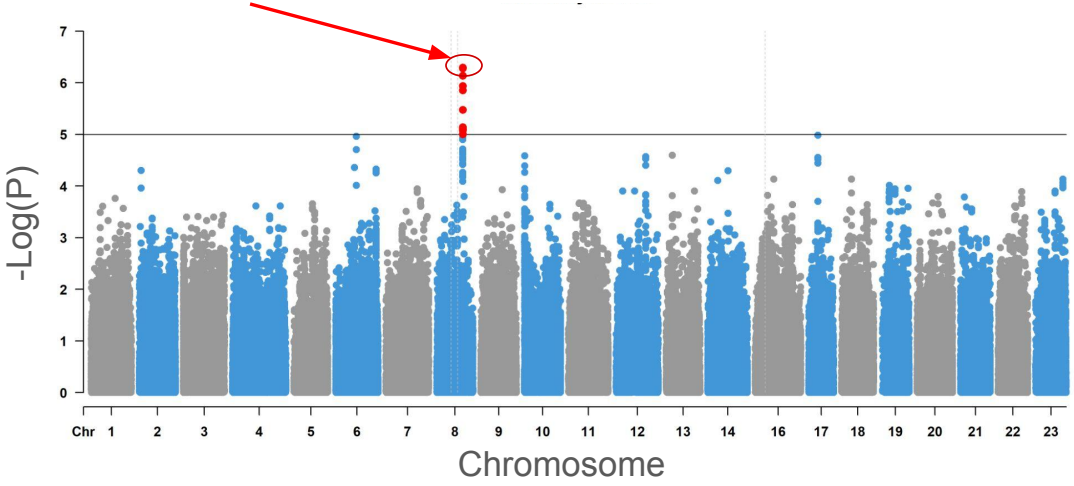


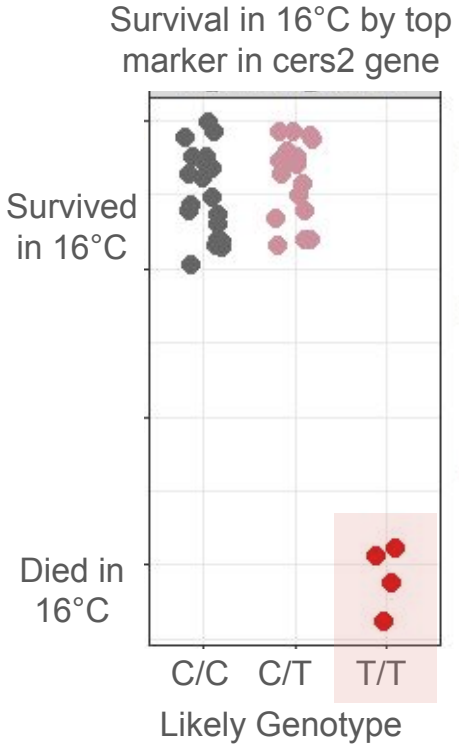
SLC274a codes for the long-chain fatty acid transport protein 4, mediates the levels of long-chain fatty acids in the cell



10 SNP markers in 1 gene, ceramide synthase 2 (cers2)

associated with **mortality in warming (16°C)** in Pacific cod juveniles







Genetic variation in cers2 associated with mortality in warming, higher cers2 expression, lower liver lipid & growth rates

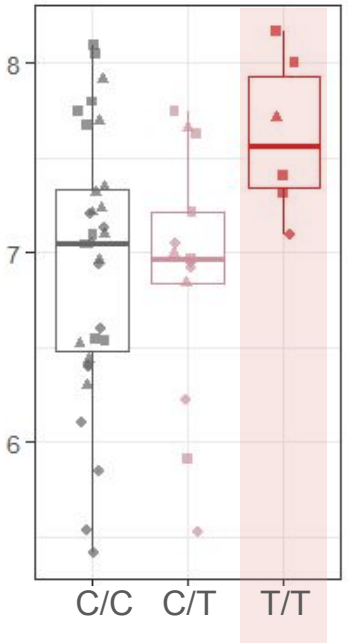
Cers2 synthesizes long-chain fatty acids (ceramides), involved in membrane structure, stress signaling

Who can survive heat waves?

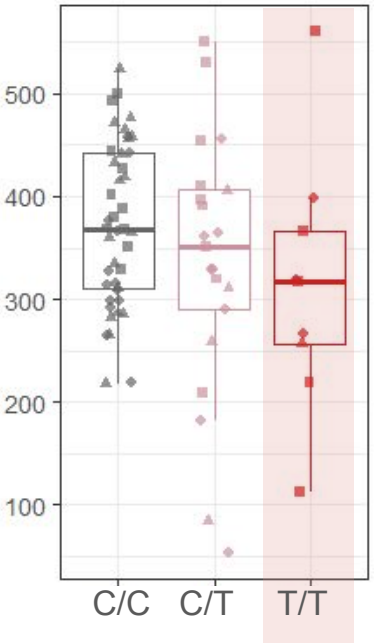
Maybe not young cod with T/T in cers2 marker

… *not necessarily causal, and traits are polygenic!!*

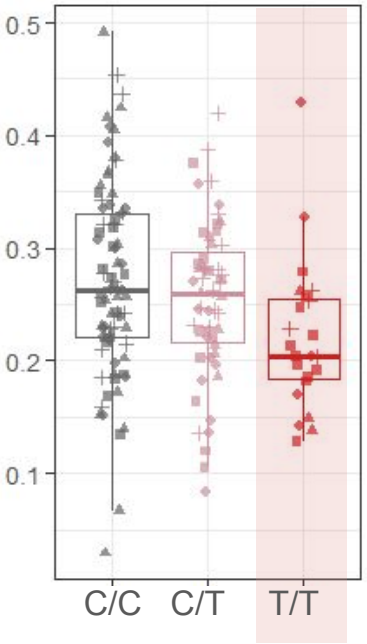
Ceramide synthase 2 expression in liver



Lipid concentration in liver



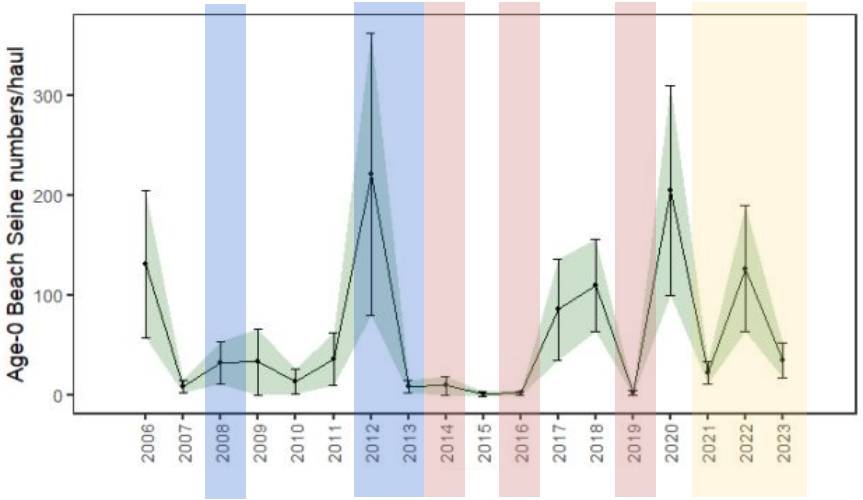
Growth rate during acclimation (all at 6°C)





Who actually survived heat waves?

WGS from archived Pacific cod juvenile (age-0)



Before Heatwave Since

Genetic Time Series from Beach Seine Survey

- Juveniles from **nine distinct years**, spanning heatwaves (2008-2023)
- ~40 per year, all in July off Kodiak (here!)
- Signals of selection for trait-associated markers?
- ***Preliminary Analysis*: track polygenic scores for fitness traits over time**



Polygenic scores, calculated from genotypes at SNPs associated with growth, liver lipid content, survival

POLYGENIC SCORE (PGS)

Combine many genetic effects into one score

1. GENOTYPE (DOSAGE)
0, 1, or 2 copies of the effect allele

SNP	You
SNP1	1
SNP2	2
SNP3	1
...	...
SNPm	0

×

2. EFFECT SIZE (β)
Effect of the allele from GWAS

SNP	β
SNP1	+0.12
SNP2	−0.08
SNP3	+0.05
...	...
SNPm	−0.03

=

3. POLYGENIC SCORE (PGS)
Sum across all SNPs

Your PGS
+0.23



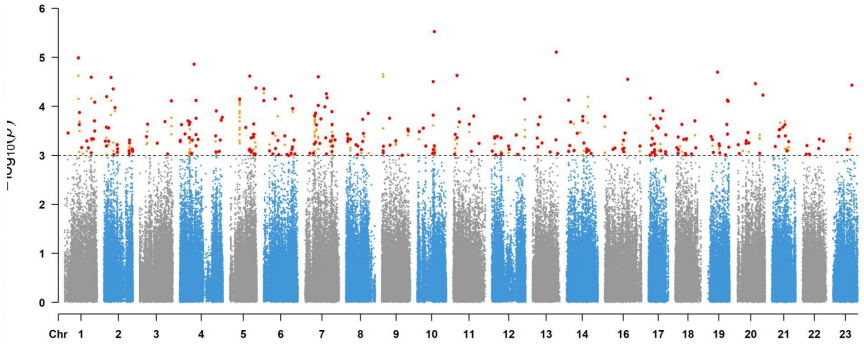
Lower PGS

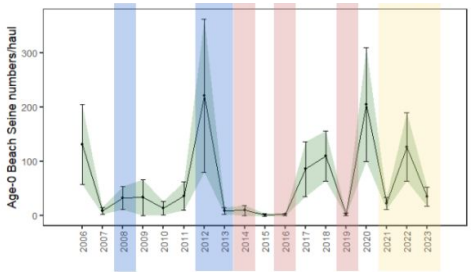
More "detrimental" alleles associated with ***lower growth, less lipid***

Higher PGS

More "beneficial" alleles associated with ***higher growth, more lipid***

Polygenic score for ***Liver Condition (HSI) uses 280 SNPs***





PGS, 40 per year

- *Growth Rates*
- *HSI*
- *Mortality*
- *Triglycerides*
- *Free Fatty Acids*
- *Sterols, Polar Lipids*



Polygenic scores for HSI, growth rate (weight) increase over time in wild-caught juveniles (*preliminary!*)

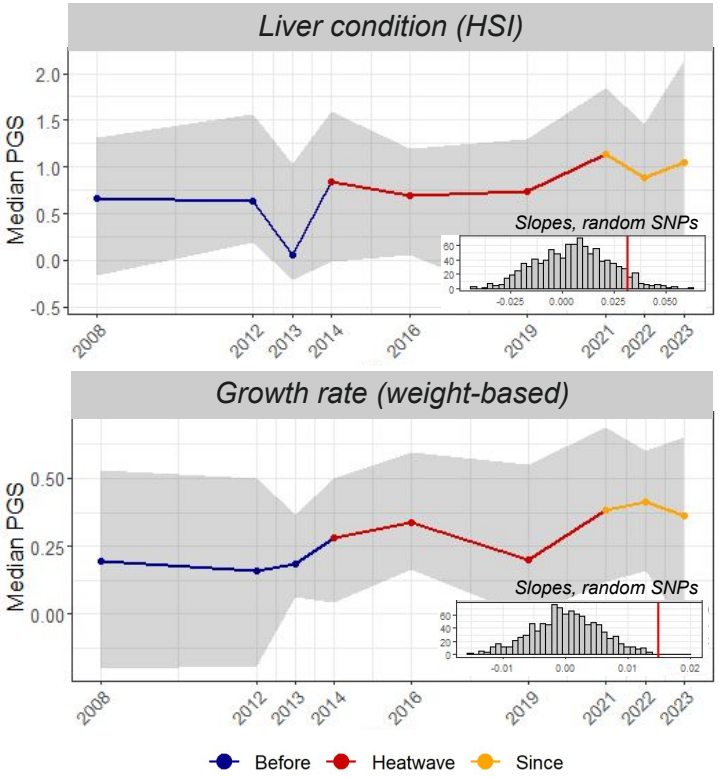
Median Polygenic Scores ± IQR

Larger livers
Faster growth

+

-

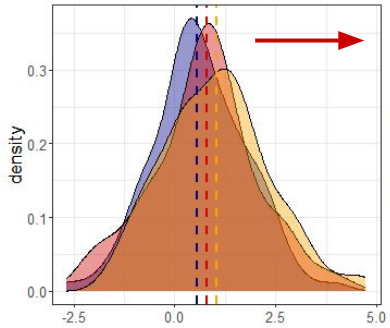
Smaller livers
Slower growth



Weak positive trend for HSI

Slope: +0.028 / year (+0.42 total)

P = 0.068, R^2: 0.016

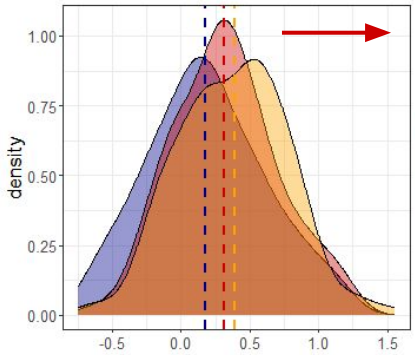


PGS for HSI

Clearer positive trend for growth rate

Slope: +0.015 / year (+0.22 total)

P = 0.0006, R^2: 0.033



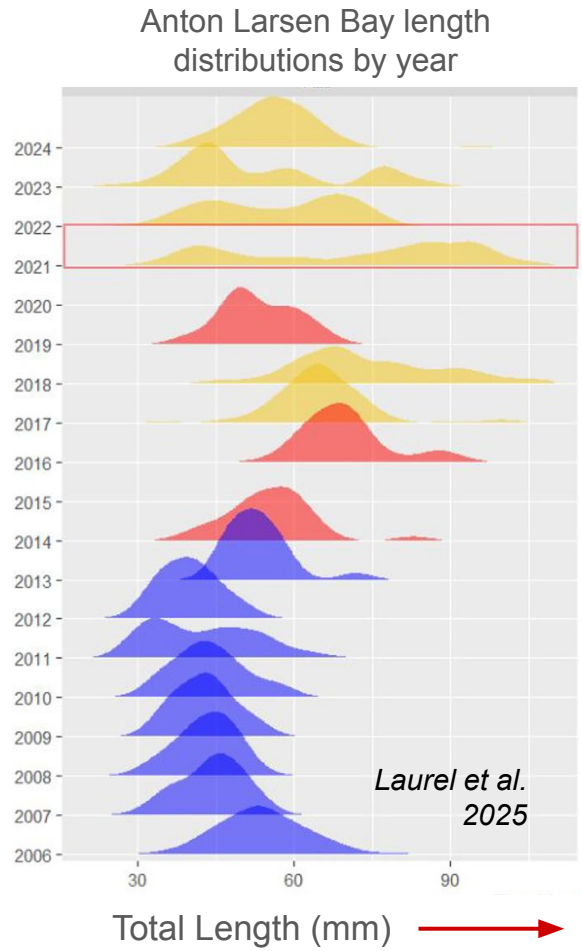
PGS for growth rate



Summary of Findings

- Experiment shows that juvenile cod **growth & lipid allocation are sensitive** to temperature.
- **Fitness traits are positively correlated** (growth rate, liver lipid allocation) - *In press, GCB-Comm*
- **Increasing Polygenic Scores** for growth rate, HSI suggest **warming may be selecting for individuals that grow faster**, more efficiently allocate energy.
- Interestingly, T_{max} (thermal optima) for growth estimated +0.6°C compared to 2012 experiment (*Laurel et al. 2016*)
- Also evidence of earlier spawning group (e.g. *Laurel et al. 2025),* and variation in migration patterns *(Spies et al. in prep)*, both genetically linked

Genetic selection + earlier spawning and/or migration may underlie shifted length distribution of Pacific cod juveniles in Anton Larsen Bay



Contributors & Funders

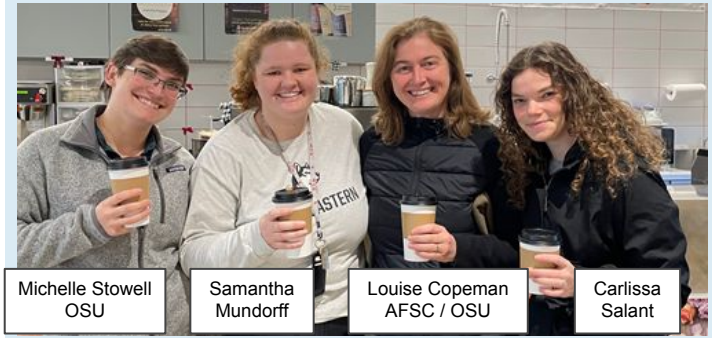
Juvenile Collection & Experimental Data


Ben Laurel, AFSC


Mary Beth Rew Hicks, AFSC


Paul Iseri, AFSC

Lipid Data


Michelle Stowell OSU

Samantha Mundorff

Louise Copeman AFSC / OSU

Carlissa Salant



‘Omics Data & Bioinformatics guidance


Ingrid Spies, AFSC


Sara Schaal, AFSC & UO


Wes Larson, AFSC


Steven Roberts, UW


Laura Timm, AFSC & UAF


Kathleen Durkin, UW

Extra slides





“Genome to Phenome” dataset across temperatures for Pacific cod





→ Individuals were tagged ←

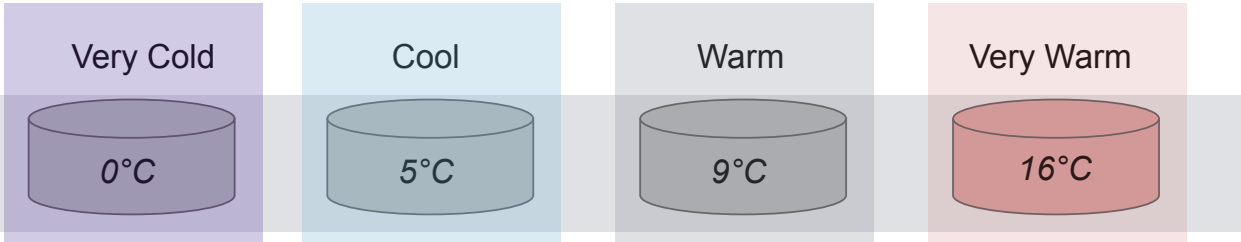
a. Genetics (lcWGS)

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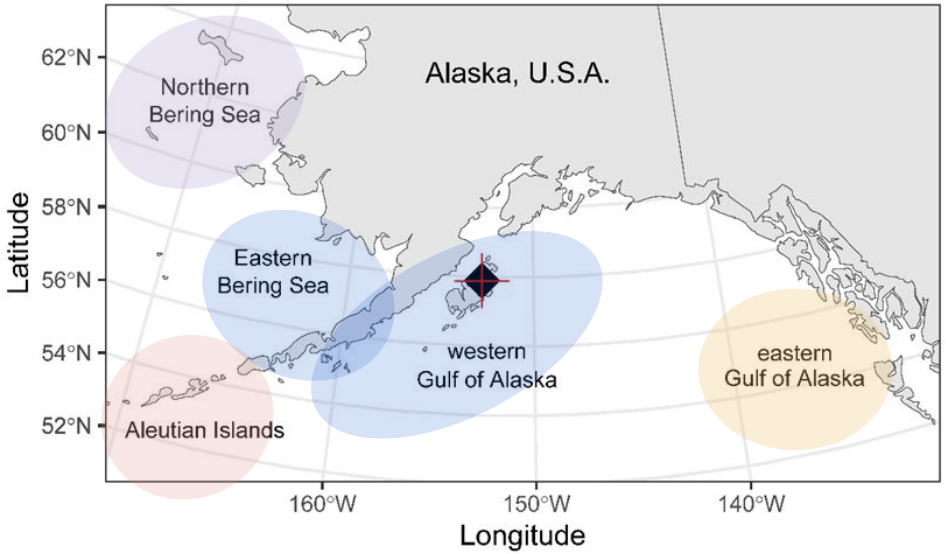
c. Fitness Traits
   - Growth rates
   - Liver & Body Condition
   - Liver lipid concentrations (n=25/temp)
   - Survival

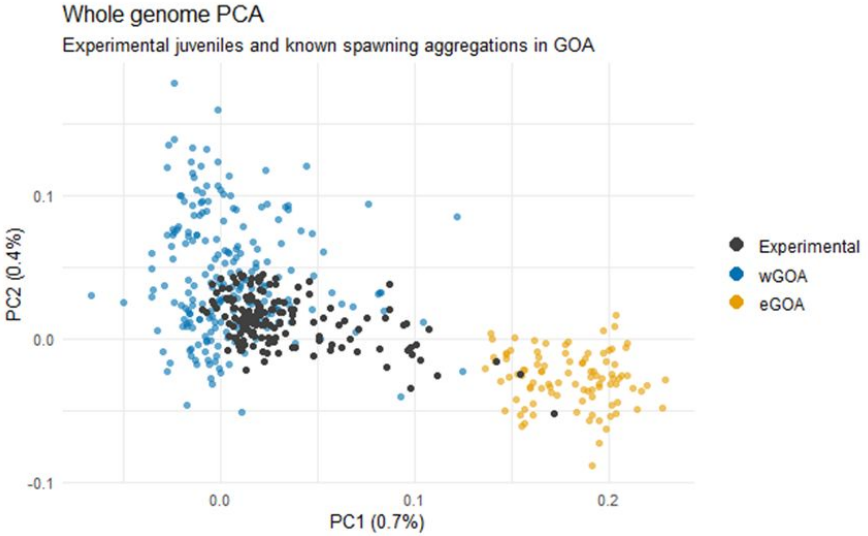


n=40 / temp.
4 week treatment



Genetic data was used to identify population represented in experiment.



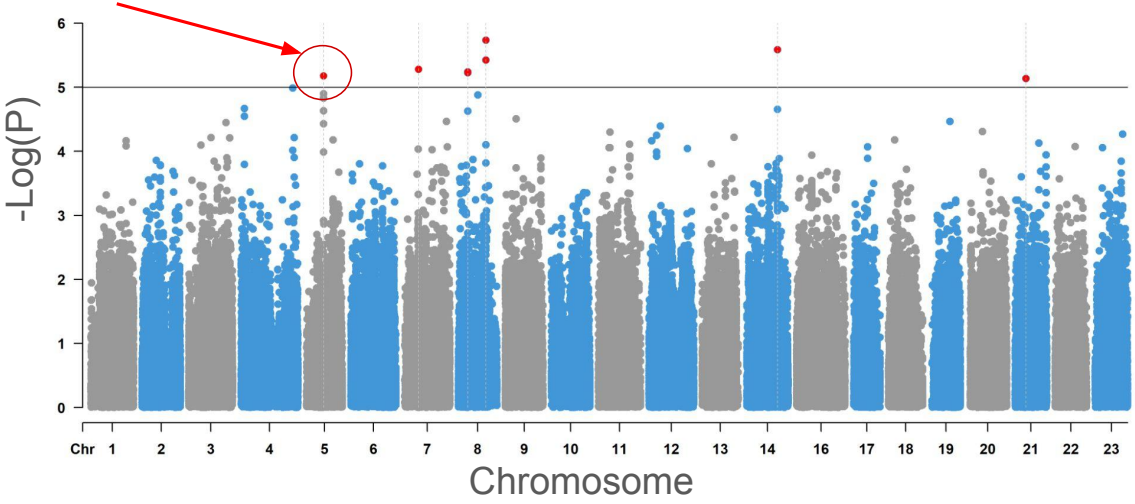


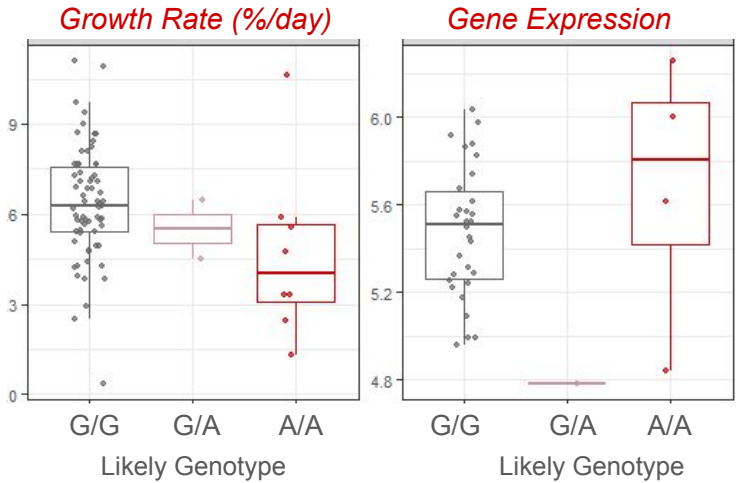
Schaal et al. 2026, https://doi.org/10.1111/eva.70174

Most juveniles from west Gulf of Alaska / EBS population, some with eastern GOA ancestry



SNP markers putatively associated with **Growth Rate** by weight in Pacific cod juveniles





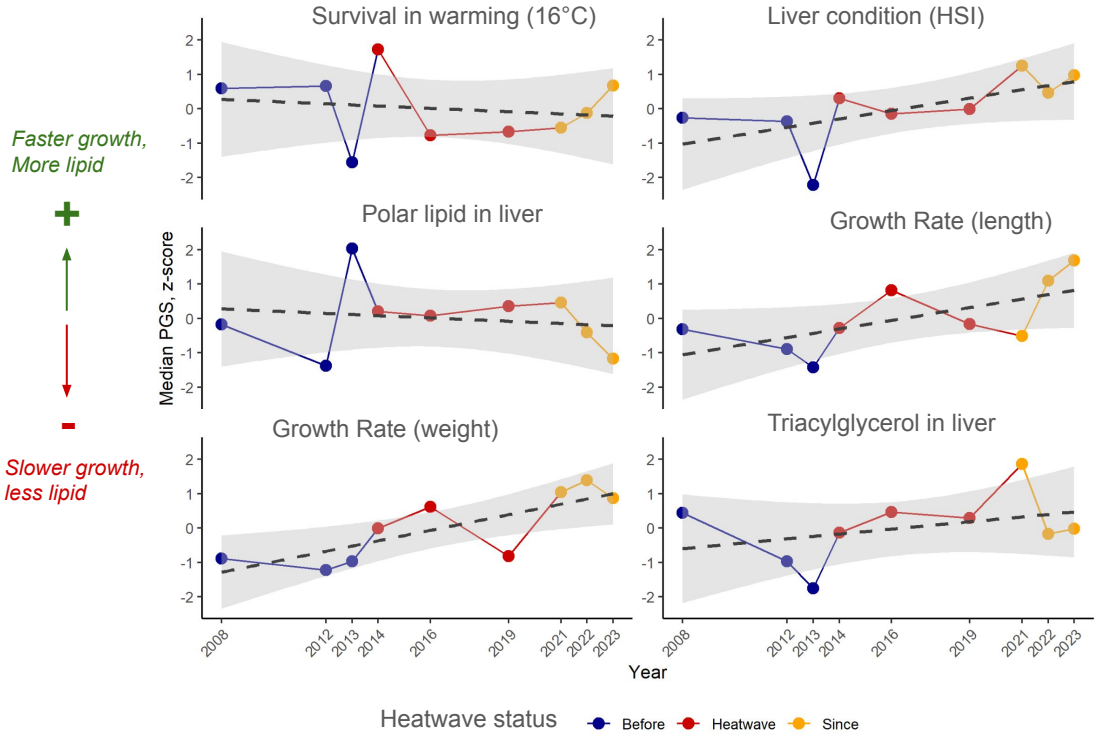
Growth rate and expression of dhx30 gene varies by genotype, more pronounced in warm temperatures

dhx30 is essential to mitochondrial ribosome assembly, and nervous system development

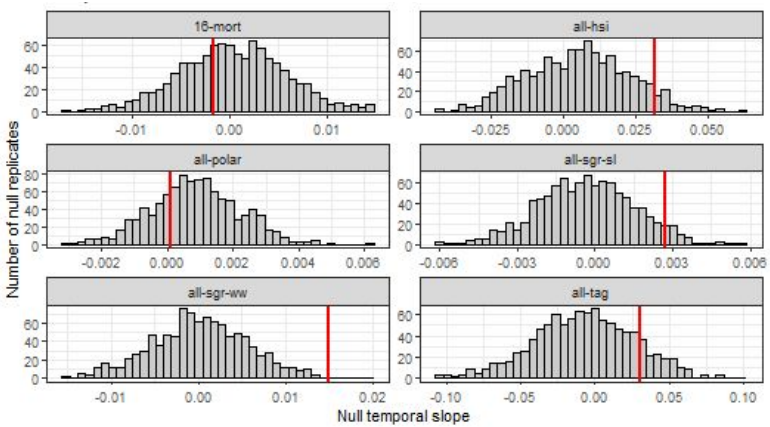


Polygenic scores, calculated from genotypes at markers associated with growth, liver lipid content

Median Polygenic Scores by trait through time

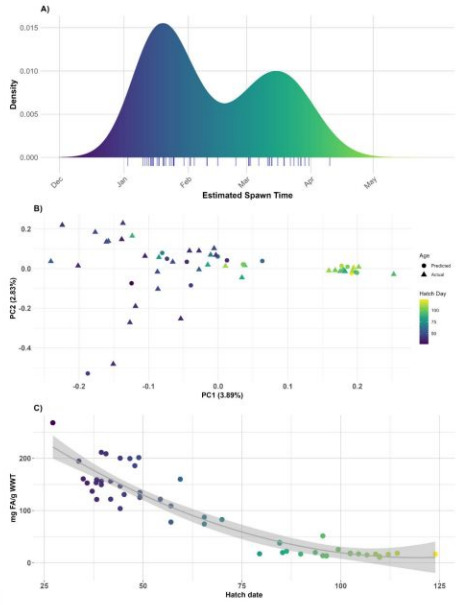


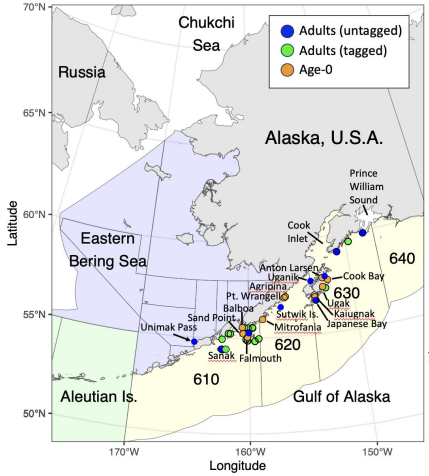
Distribution of slopes fit to Polygenic Score using random SNPs



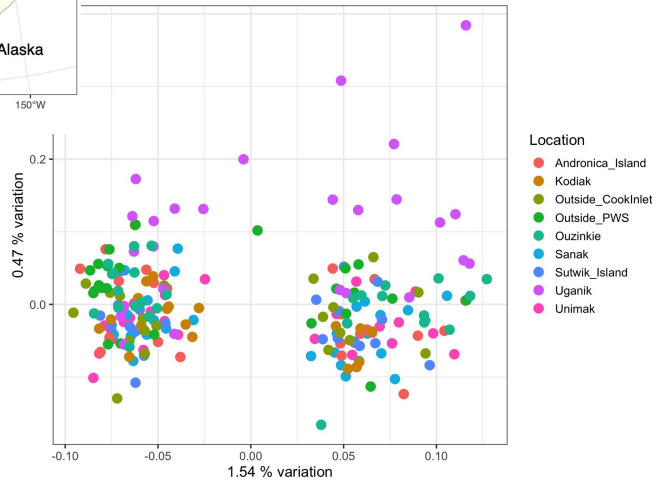


There is variation within wGOA/EBS Pacific cod, which is good!





New research! *There are multiple genetically distinct groups in GOA with varying spawn timing, depth (and possibly temperature) preferences - "ecotypes"*



First evidence of genetically distinct winter- and spring-spawning components of Pacific cod in the Gulf of Alaska

Authors: Benjamin Laurel, Jessica Miller, Louise Copeman, Alisa Abookire, Laura Spencer, Sara Schaal, Mary Beth Rew Hicks, Nicholas Strait, Lauren Rogers, and Ingrid Spies | AUTHORS INFO & AFFILIATIONS

Publication: Canadian Journal of Fisheries and Aquatic Sciences • 16 January 2026 • https://doi-org.offcampus.lib.washington.edu/10.1139/cjfas-2025-0168